

Data Analysis Exercise #1 Recap

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Why we use R?

- Advantage of using R
 - Open Source & Free
 - Designed for statistics and data analysis
 - Reproducible research
 - Statistics & Data Science, Social Sciences, Economics, and Public Health

Encouraging notes:

- Everyone struggles at first (learning syntax)
- Error messages are part of learning
- Use “help()” in R
- Google/GPT is a good mentor

What is RStudio?

- RStudio makes R easier to use (Dashboard)
- Installation
 - From: <https://posit.co/download/rstudio-desktop/>

Windows 10/11	<u>RSTUDIO-2026.01.0-392.EXE</u> ↓	314.17 MB	<u>525403C0</u>
macOS 14/15/26	<u>RSTUDIO-2026.01.0-392.DMG</u> ↓	670.03 MB	<u>EE3F39BB</u>

Source Pane:

Edit and run scripts.

Your code will not be evaluated until you “Run” them to the console.

Console Pane:

R commands run are shown here, and non-graphic output and errors are displayed.

The screenshot displays the RStudio interface with four main panes:

- Source Pane:** Contains R code for generating histograms. A red arrow points from the 'Run' button to the console.
- Environment Pane:** Shows the global environment with objects like 'dat' (2000 obs. of 20 variables) and 'Table1a'. A red arrow points from the 'Values' column to the console.
- Console Pane:** Displays the output of the R commands, including a list of education levels and their frequencies. A green arrow points from the console to the source pane.
- Plots Pane:** Shows a histogram titled 'Histogram of dat\$X1MTHID' with frequency on the y-axis and X1MTHID on the x-axis. A purple arrow points from the plot to the console.

```
# file will be saved in location specified above (working directory)
write.csv(Table2, "Ex1-Table2.csv", row.names=TRUE, na="")

# generate histogram of 9th grade mathematics self-efficacy score (X1MTHID)
# file will be saved in location specified above (working directory)
png("Ex1-Figure1.png", width=6.5, height=4, units="in", res=600)
par(mar=c(3,3,1,1), mgp=c(2,0.5,0), las=1, tcl=-0.2) # graphical parameters; see help
(par) for details
hist(dat$X1MTHID, xlab="Math Identity Score", main="")
dev.off()

# generate histogram of 9th grade mathematics self-efficacy score (X1MTHID)
# file will be saved in location specified above (working directory)
png("Ex1-Figure2.png", width=6.5, height=4, units="in", res=600)
par(mar=c(3,3,1,1), mgp=c(2,0.5,0), las=1, tcl=-0.2) # graphical parameters; see help
(par) for details
hist(dat$X1MTHID, xlab="X1MTHID", col = "lightblue")
```

Console output:

```
(01) Less than high school          (02) High school diploma or GED          233
      5
(03) Start an Associate's degree    (04) Complete an Associate's degree    118
      12
(05) Start a Bachelor's degree      (06) Complete a Bachelor's degree    309
      8
(07) Start a Master's degree         (08) Complete a Master's degree    349
      17
(09) Start Ph.D/M.D/Law/other prof degree (10) Complete Ph.D/M.D/Law/other prof degree 360
      17
      (11) Don't know                <NA>
      398                            174

> hist(dat$X1MTHID, xlab = "X1MTHID", col = "lightblue")
>
```

Environment Pane:

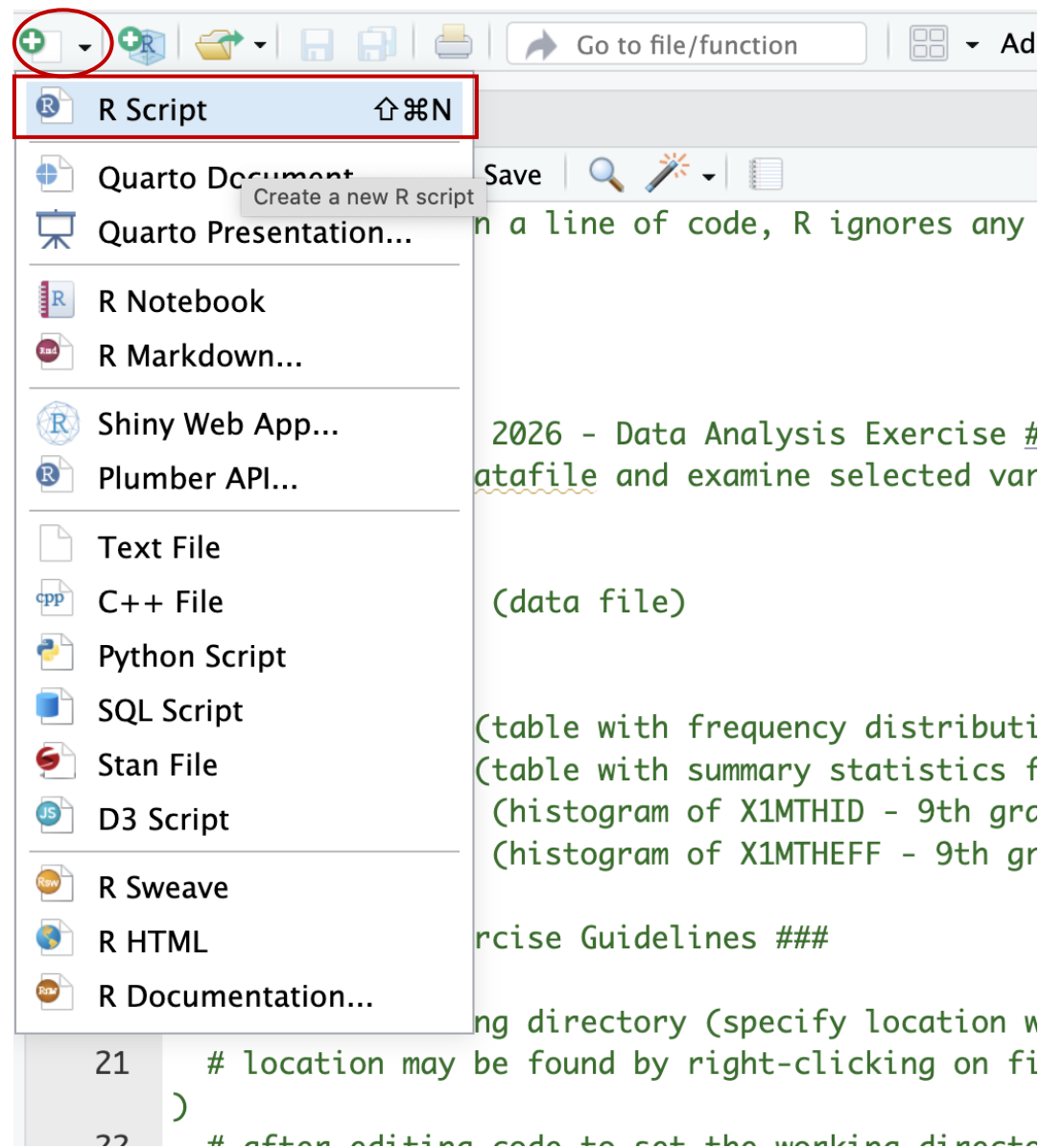
Overview of objects (dataset, values, parameters) you have created.

Plots, Packages, and Help Pane:

Commonly used to view graphics, install packages, and view help.

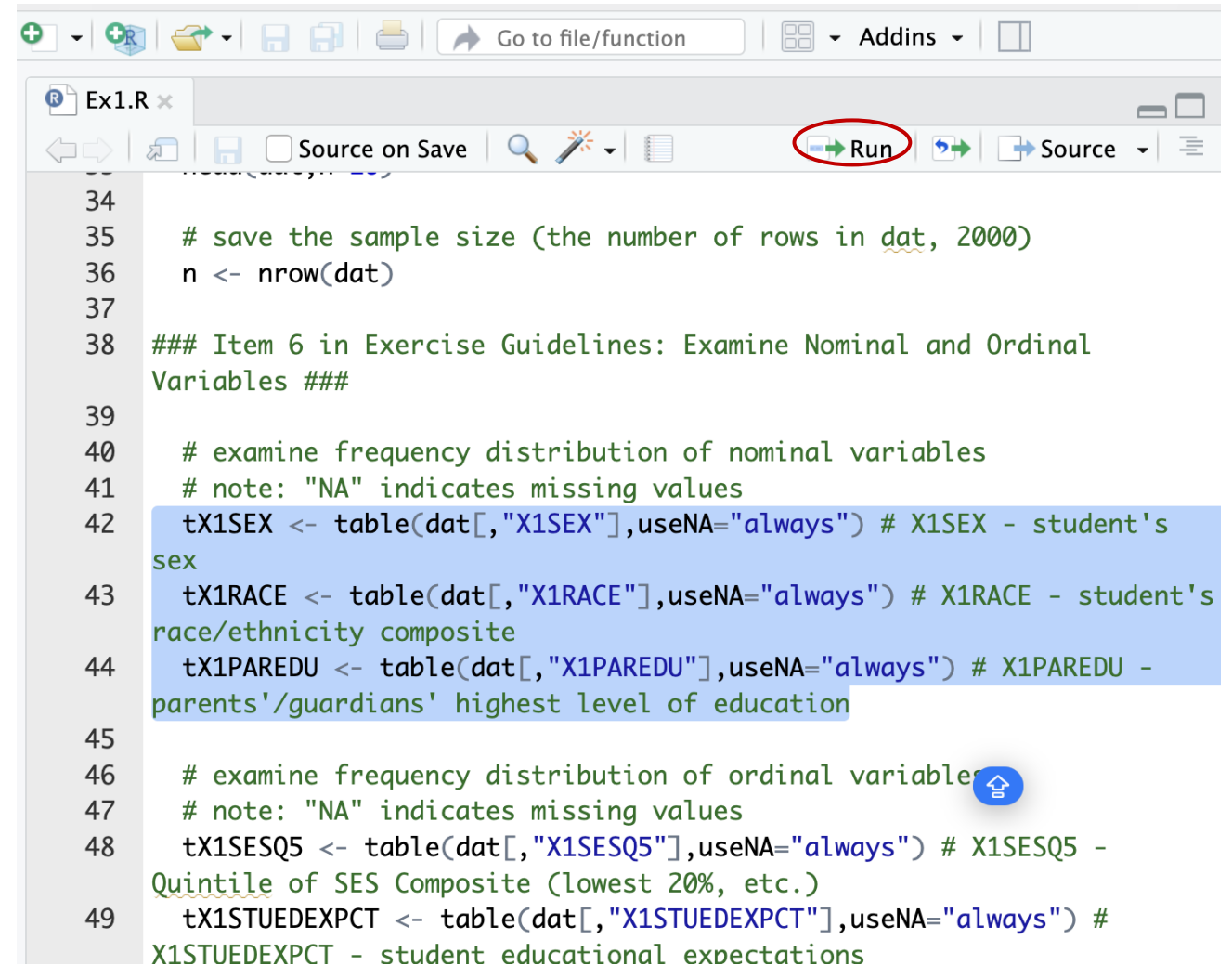
Open R Script

- Double click the .R file
- Or open it from RStudio



Run R Script

- Highlight the lines + click “Run” button
- Shortcut:
 - Specific lines:
Ctrl+Enter (Windows/Linux)
Cmd+Return (macOS)
 - Multiple lines:
Ctrl+Shift+Enter (Windows/Linux)
Cmd+Shift+Return (macOS)



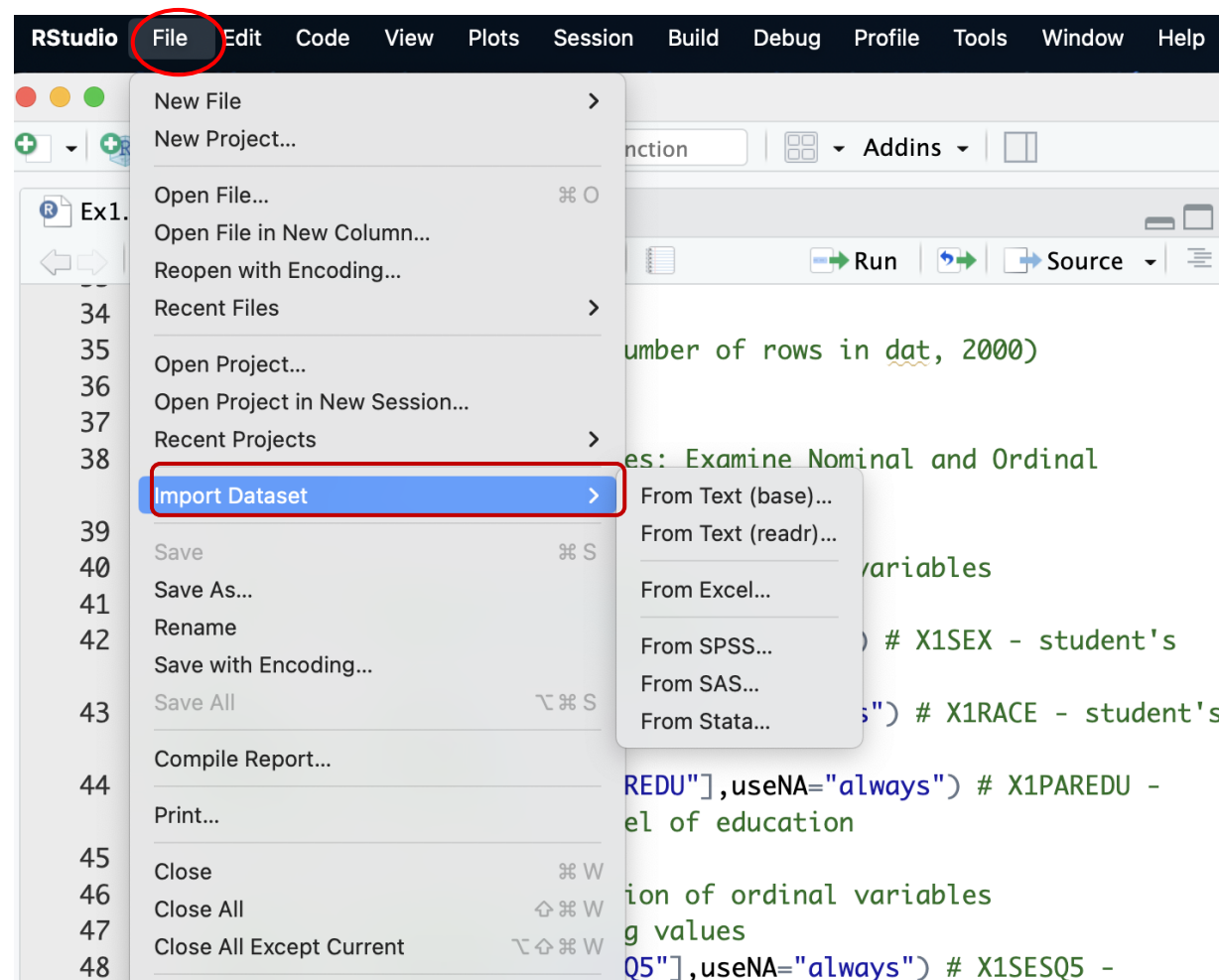
```
34  
35 # save the sample size (the number of rows in dat, 2000)  
36 n <- nrow(dat)  
37  
38 ### Item 6 in Exercise Guidelines: Examine Nominal and Ordinal  
39 Variables ###  
40 # examine frequency distribution of nominal variables  
41 # note: "NA" indicates missing values  
42 tX1SEX <- table(dat[, "X1SEX"], useNA="always") # X1SEX - student's  
sex  
43 tX1RACE <- table(dat[, "X1RACE"], useNA="always") # X1RACE - student's  
race/ethnicity composite  
44 tX1PAREDU <- table(dat[, "X1PAREDU"], useNA="always") # X1PAREDU -  
parents'/guardians' highest level of education  
45  
46 # examine frequency distribution of ordinal variables  
47 # note: "NA" indicates missing values  
48 tX1SESQ5 <- table(dat[, "X1SESQ5"], useNA="always") # X1SESQ5 -  
Quintile of SES Composite (lowest 20%, etc.)  
49 tX1STUEDEXPCT <- table(dat[, "X1STUEDEXPCT"], useNA="always") #  
X1STUEDEXPCT - student educational expectations
```

How to import data into R?

From the tool bar

Step1: “File”

Step2: “Import Dataset



How to import data into R?

- From the data import code “read.csv()”

(“) and (‘) is interchangeable

- For macOS

Copy & Paste your file path

```
dat <- read.csv("/Users/username/Desktop/Dataset/hsls0913_hw.csv")
```

Shortcut: **Right-click** the file, hold down the **Option** key, and the "Copy" option will change to "**Copy ... as Pathname**".

- For windows

Replace \ with / or \\

```
dat <- read.csv("C://your file path name/hsls0913_hw.csv")
```

Shortcut: Hold down the **Shift** key and **right-click** on the file, then select "**Copy as path**" from the context menu

Basic functions for descriptive statistics

- Recap for the code

```
# examine frequency distribution of nominal variables
```

```
# note: "NA" indicates missing values
```

```
tX1SEX <- table(dat[, "X1SEX"], useNA="always") # X1SEX - student's sex
```

“**table()**” counts how many times each value appears.
Used for **categorical / ordinal variables**

In R, we have “**dat[rows, columns]**”, “,” means keep all rows, “**X1SEX**” means **only** select the column named “X1SEX”.
“**dat[1:10,]**” means only select first 10 rows, and remain all the columns.

Tell R how to handle missing values:
= “**always**” means Always show NA
= “**if any**” means Show NA if they exist
= “**no**” (default) means Ignore NA

Basic functions for descriptive statistics

“**rbind()**” means **Row Bind**. Row combine what we have here.

“**c()**” means **combine**. Create a tow containing two text strings “n” and “%”.

“**cbind()**” means **Column Bind**. It takes two or more objects and sticks them together side-by-side as columns.

Calculates the percentages

```
Table1a <- rbind(c("n", "%"), cbind(tX1SEX, 100*tX1SEX/n)) # add %
```

```
rownames(Table1a)[1] <- "tX1SEX" # add variable name to table
```

Together we “<-” assign this objects as our “Table1a”.

“**rownames()**” means retrieves a list of the labels attached to “Table1a” rows. And “[1]” is an index that tells R to focus on the first item.

```
> Table1a
      tX1SEX
(1) Male  "1031" "51.55"
(2) Female "967"  "48.35"
<NA>      "2"   "0.1"
```

⇒

```
      tX1SEX
(1) Male  "1031" "51.55"
(2) Female "967"  "48.35"
<NA>      "2"   "0.1"
```

Basic functions for descriptive statistics

```
# create subtable for X1MTHID (mathematics identity)
Table2a <- c("X1MTHID", # variable name
sum(!is.na(dat[, "X1MTHID"])), # count number of records for which variable is observed (not Missing)
sum(is.na(dat[, "X1MTHID"])), # count number of records for which variable is NA (Missing)
quantile(dat[, "X1MTHID"], probs=c(0, 0.25, 0.50, 0.75, 1.00), na.rm=TRUE), # min, quartiles, max
mean(dat[, "X1MTHID"], na.rm=TRUE), # obtain mean ("na.rm=TRUE" lets NA values be ignored)
sd(dat[, "X1MTHID"], na.rm=TRUE), # compute standard deviation ("na.rm=TRUE" lets NA values be ignored
NULL))
```

Create a summary list (vector) for the “X1MTHID”. “**c()**”: As before, this stands for **Combine**. It collects all the individual calculations inside the parentheses **()** and separated by “,”

Basic functions for descriptive statistics

“**png()**” means basic setup. You file name, your physical size of the image, the size is in inches, and resolution is 600 (suitable for academic publication).

```
# file will be saved in location specified above (working directory)
png("Ex1-Figure1.png",width=6.5,height=4,units="in",res=600)
par(mar=c(3,3,1,1),mgp=c(2,0.5,0),las=1,tcl=-0.2) # graphical parameters; see help(par)
hist(dat$X1MTHID,xlab="Math Identity Score",main="")
dev.off()
```

“**par(...)**” stands for **parameters**.

mar=c(3,3,1,1): Margins. The order is Bottom, Left, Top, Right.

mgp=c(2,0.5,0): Margin Placement.

las=1: Label Axis Style (numbers be **horizontal**)

tcl=-0.2: Tick Length. Makes the little tick marks on the axis shorter

Where you actually draw the plot.

dat\$X1MTHID: The column of data to plot.

xlab="": The label text for the X-axis.

main="": The main title.

May you have a great adventure in R!

